

# GENERAL THETA FOUNDATIONS I: WORD PROFILES AND ARITHMETIC FLUCTUATIONS OF NUMERICAL MEMORY

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ABSTRACT. We associate two finite-scale quantities to a finite orthogonal command alphabet: an optimized word-packet distortion and a compatible convex-enclosure contraction. The first bounds every hidden stochastic realization through a nonuniform interval budget; the second constructs exact realizations with common command rows. Neither requires a spectral gap. In the planar case, optimal packet distortion has a finite second-order cone formulation. These profiles yield a numerical-memory exponent  $r/(2r + 1)$  for every  $r$ -angle alphabet of minimal ordinary dual Diophantine type, a class of full Lebesgue measure; explicit logarithmic bounds hold almost everywhere. For every irrational single angle we identify a sequence of horizons on which width has cubic-root order. For every Liouville angle the logarithmic width exponent has lower limit zero and upper limit at least one third, so a single power law need not exist. All lower bounds allow arbitrary hidden states and time-dependent stochastic rows, and the upper constructions preserve each numerical probability exactly. The resource is nonuniform clocked atomic-row label width, with a fixed signal, a fixed sufficiently small wordwise error, and one selected terminal query.

## 1. INTRODUCTION

A finite-dimensional orbit may admit a small positive factorization at each time and still require a growing register when those factorizations must be executed by one causal machine. For a fixed alphabet, that growth also depends on the distribution of its finite words. A qualitative assertion that all spectral gaps vanish loses precisely the arithmetic information which distinguishes such examples.

We study the numerical response

$$(1.1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\},$$

where the command word  $w$  has prescribed length  $N$ , the seed  $x$  is a unit vector, and only one final coordinate  $j$  is queried. The matrices in  $U_w$  are orthogonal. A classical register must preserve these probabilities, or approximate every one of them within a fixed binary total-variation error. Its stochastic rows may vary arbitrarily with the epoch. Section 2 defines the resource without a uniformity or finite-random-bit assumption.

The first main result is a pair of operational bounds in terms of finite data. For words of length  $B$ , let  $\Gamma_{\mathcal{A}}(k, B)$  be the largest, over word distributions, of the smallest spherical distortion after replacing the word orbit by  $k$  centroid directions. This definition optimizes the converse law and permits every initial direction. Theorem 3.3 charges  $-\log(1 - \Gamma_{\mathcal{A}}(K_t, B))$  to any interval of length  $B$  ending at time  $t$ , even when all other registers are unrestricted. A finite interval recurrence combines all such charges and bounds actual final error. In the opposite direction, Theorem 3.5 turns a finite polytope and its worst command contraction into common stochastic rows. These are distinct lower and upper profiles; we do not assume that an optimal hidden realization must be a vector-state polytope.

For circle actions, Theorem 4.1 removes the continuous search over centroid directions. The exact optimized packet value is a finite second-order cone problem indexed by cyclic partitions of the word phases. Thus a finite packet certificate is computationally different from an infinite harmonic-gap assertion. The theorem is not an efficiency assertion in a succinct word length.

Our arithmetic conclusions go beyond uniform bad approximation. For  $\alpha \in \mathbb{R}^r$  with  $1, \alpha_1, \dots, \alpha_r$  rationally independent, write

$$(1.2) \quad \omega^*(\alpha) = \limsup_{\|m\|_1 \rightarrow \infty} \frac{-\log \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}}{\log \|m\|_1}.$$

The alphabet consists of the identity and the rotations through  $2\pi\alpha_i$ . Throughout,  $r$ , the alphabet,  $0 < \rho \leq 1/10$  and  $0 \leq \varepsilon < \rho/(2\sqrt{2})$  are fixed.

**Theorem 1.1** (Minimal Diophantine type). *If  $\omega^*(\alpha) = r$ , then*

$$(1.3) \quad \lim_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = \frac{r}{2r+1}.$$

*For Lebesgue-almost every  $\alpha$ , and every fixed  $\delta > 0$ , there are positive constants  $c, C$  such that, for  $N \geq 2$ ,*

$$(1.4) \quad c \frac{N^{r/(2r+1)}}{(\log(N+2))^{2r(1+\delta)/(2r+1)}} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)} (\log(N+2))^{2r^2(1+\delta)/(2r+1)}.$$

*The upper inequalities are achieved by exact common-row machines.*

The exponent in (1.3) need not imply a two-sided constant-factor law. The earlier dual badly approximable theorem supplies the stronger  $\Theta$  conclusion on its stated class, and is recovered in Corollary 5.4. Theorem 1.1 instead allows arbitrarily small Diophantine constants along subsequences. The quantitative arithmetic ingredients are classical; the memory conclusions concern their compatible realization and all-hidden-state converse.

There is also a genuine failure of a single exponent, even for one fixed irrational angle.

**Theorem 1.2** (Irrational horizons and Liouville fluctuations). *For every irrational  $\alpha$  there are horizons  $M_n \rightarrow \infty$  such that  $W_{M_n,\varepsilon}(\alpha) = \Theta(M_n^{1/3})$ . If  $\alpha$  is Liouville, then*

$$(1.5) \quad \liminf_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = 0, \quad \frac{1}{3} \leq \limsup_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} \leq \frac{1}{2}.$$

*Nevertheless  $W_{N,\varepsilon}(\alpha) \rightarrow \infty$ . All displayed upper bounds have exact realizations.*

The upper limit in (1.5) is not determined here. The result establishes oscillation between subpower subsequences and a positive-power subsequence; it does not replace that distinction by a guessed exponent for all irrationals.

Relation to previous work. Finite codebooks, nearest-center partitions and positive convex realizations are classical tools [5, 11]. The endpoint conditional-centroid contraction used below was established in the preceding packet development [7, 8]; we give its complete proof and use it to define a finite alphabet profile. The general interval optimization, the circle word-law program, and the metric and Liouville memory conclusions are the new statements of this revision. Classical Dirichlet approximation, continued fractions and algebraic examples are not counted as new arithmetic. The direct comparison with Dick–Goda–Larcher–Pillichshammer–Suzuki [4], including their Theorem 3.9 and Lemma 3.11, is made in Section 8.

| Quantity                             | Meaning                                                              |
|--------------------------------------|----------------------------------------------------------------------|
| $K_t, W_{N,\varepsilon}$             | Available labels at a cut; minimum whole-run peak.                   |
| $\Gamma_{\mathcal{A}}(k, B)$         | Best finite-word distortion against $k$ endpoint directions.         |
| $\mathcal{I}_N(K_{\bullet})$         | Maximum sum of distortion charges over disjoint intervals.           |
| $\lambda_{\mathcal{A}}(k, a)$        | Least common enclosure dilation with $k$ vertices and inradius $a$ . |
| $\psi_{\alpha}(H), \eta_{\alpha}(q)$ | Dual separation and simultaneous rational-approximation errors.      |

## 2. THE NUMERICAL EXPERIMENT AND THE REGISTER

Let  $V = \mathbb{R}^D$ , let  $\mathcal{A}$  be a nonempty finite alphabet, and fix orthogonal matrices  $U_a$ ,  $a \in \mathcal{A}$ . Write  $U_w = U_{a_N} \cdots U_{a_1}$  in the order of execution. Let  $X \subset \mathbb{S}^{D-1}$  be a nonempty finite seed set and  $0 < \rho < 1$ . The required law is (1.1). Commands are externally supplied, not chosen by a policy inside the simulator. No distribution on command words is part of the specification.

**Definition 2.1** (Clocked atomic-row realization). At horizon  $N$ , a machine consists of finite sets  $S_t$  with  $|S_t| = K_t$  for  $0 \leq t \leq N$ , an initialization probability row  $E_x$  on  $S_0$ , stochastic matrices  $T_{t,a} : S_{t-1} \rightarrow S_t$ , and terminal columns  $d_j : S_N \rightarrow [-1, 1]$ . Its numerical mean is

$$E_x T_{1,a_1} \cdots T_{N,a_N} d_j.$$

After the atomic query  $j$ , it draws and holds the binary answer with that mean. The final answer cut has two labels. Its peak is  $\max(2, K_0, \dots, K_N)$ . Available labels count even when unreachable; adding unused normalized rows is permitted.

All persistent randomness, input history and protocol distinctions must be in the current label. Only the external epoch, the prescribed horizon and the read-only row tables are free. Table construction, lookup, arithmetic and exact sampling of known real stochastic rows are uncharged. Matrices, rows and state sets may depend on  $N$ ; this is nonuniform positive-realization width, not uniform computational space. A common-row construction uses the same command matrices at all command epochs but may still have an  $N$ -dependent decoder. It need not answer at earlier stopping lengths.

Let  $W_{N,\varepsilon}$  be the least peak among machines whose every binary row differs from (1.1) by at most  $\varepsilon$  in total variation. For fixed finite  $K_{\bullet}$  the optimal actual error  $\mathcal{E}_N(K_{\bullet})$  is attained: the row parameters lie in a finite product of compact sets and the error is a finite maximum of continuous functions. For planar results we use  $X = \{\pm e_1, \pm e_2\}$  and include the identity letter. We keep  $\rho \leq 1/10$  and  $\varepsilon < \rho/(2\sqrt{2})$ .

**Lemma 2.2** (Calibration under any external test law). *Fix a seed  $x \in X$  and any law on the complete command word, independent of private initialization. Let  $Y_t = U_{a_t} \cdots U_{a_1} x$ ,  $Z_t = \mathbb{E}[Y_t | S_t]$  and  $q_t = \mathbb{E}\|Z_t\|$ . Every machine with row error at most  $\varepsilon$  satisfies*

$$(2.1) \quad q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon.$$

*At a boundary preceding an independent complete packet, the next boundary admits the conditional-mean identity used in Lemma 3.2. No single-step independence inside a packet is asserted.*

*Proof.* For a fixed word, every coordinate of  $\mathbb{E}[d(S_N) | w] - \rho Y_N$  has modulus at most  $2\varepsilon$ . Since  $\|Y_N\| = 1$ , averaging gives

$$\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\varepsilon\sqrt{D}.$$

On the other hand this pairing equals  $\mathbb{E}\langle Z_N, d(S_N) \rangle$  and is at most  $\sqrt{D}q_N$ . This proves (2.1). Combining terminal columns is a proof device, not a requirement to provide their joint output distribution.  $\square$

### 3. FINITE WORD PROFILES AND COMPATIBLE ENCLOSURES

The results in this section apply to every finite orthogonal alphabet, including noncommuting alphabets. No connectedness, arithmetic condition or spectral gap is assumed. Invariant directions may make a converse profile zero; this is a property of the stated profile, not an omitted hypothesis.

**3.1. An optimized distortion.** For  $B, k \geq 1$  and a probability vector  $p$  on the actual words  $\mathcal{A}^B$ , define

$$(3.1) \quad D_k(p) = 1 - \max_{\substack{u \in \mathbb{S}^{D-1} \\ v_1, \dots, v_k \in \mathbb{S}^{D-1}}} \sum_{w \in \mathcal{A}^B} p_w \max_{1 \leq j \leq k} \langle v_j, U_w u \rangle,$$

$$(3.2) \quad \Gamma_{\mathcal{A}}(k, B) = \max_{p \in \Delta(\mathcal{A}^B)} D_k(p), \quad \chi_{\mathcal{A}}(k, B) = -\log(1 - \Gamma_{\mathcal{A}}(k, B)).$$

We allow  $\chi = +\infty$ . Repeated product matrices can be merged for the optimization: their probability masses add. Their representative words must still be executed one ordinary command at a time.

**Proposition 3.1** (Finite-data properties). *The maxima in (3.1)–(3.2) are attained,  $0 \leq \Gamma_{\mathcal{A}}(k, B) \leq 1$ , and the profile is invariant under simultaneous orthogonal changes of coordinates. It is nonincreasing in  $k$  and unchanged by duplicating a letter. For two indexed orthogonal alphabets  $U, V$ ,*

$$(3.3) \quad |\Gamma_U(k, B) - \Gamma_V(k, B)| \leq B \max_a \|U_a - V_a\|_{\text{op}}.$$

*For a fully listed finite alphabet of real-algebraic matrices and fixed integers  $k, B, D$ ,  $\Gamma_{\mathcal{A}}(k, B)$  and an optimizing word law are real-algebraic and can be determined by real quantifier elimination.*

*Proof.* All parameter spaces are compact and the objective is continuous. Its maximum over directions is convex in  $p$ , so  $D_k$  is continuous and concave. The maximum scalar product is at most one. By choosing a single direction parallel to  $\sum p_w U_w u$ , the inner maximum is nonnegative; hence  $D_k \in [0, 1]$ . Conjugating all directions proves invariance. Additional centers only increase that inner maximum. Product telescoping gives  $\|U_w - V_w\|_{\text{op}} \leq B \max_a \|U_a - V_a\|_{\text{op}}$ , and maxima and minima preserve this uniform bound.

For the last assertion, enumerate the product matrices. Unit-sphere equations, simplex inequalities and finitely many maximum operations define a compact semialgebraic optimization with algebraic coefficients. Maximum operations can be expressed by finite alternatives or auxiliary variables. Quantifier elimination over the real closed field of real-algebraic numbers determines the endpoint of the feasible value interval and a witness [1]. The word table may have  $|\mathcal{A}|^B$  entries; this is not a polynomial-time assertion or a decision procedure for an infinite spectral gap. No logarithm needs to be eliminated: finite budget comparisons can instead use products of  $1 - \Gamma$ .  $\square$

**Lemma 3.2** (Endpoint contraction; inherited mechanism). *Let a complete random word  $W$  with law  $p$  be independent of all information before its start. Internal letters of  $W$  may be correlated. If the endpoint register has at most  $k$  labels, then*

$$(3.4) \quad q' \leq (1 - D_k(p))q.$$

*There is no bound on the intermediate registers.*

*Proof.* Write  $z_s = \mathbb{E}[Y \mid S = s]$  at the start. Composing all internal stochastic rows yields a common endpoint kernel  $T(j \mid s, w)$ . Since the entire word and the machine's fresh randomness are independent of the past,

$$z'_j = \mathbb{E}[U_W z_S \mid S' = j].$$

This is an endpoint identity; it would not generally hold after conditioning on a correlated next letter inside the packet. Choose  $v_j = z'_j / \|z'_j\|$  for nonzero  $z'_j$ , and any unit vector otherwise. Then

$$\begin{aligned} q' &= \sum_s \mathbb{P}(S = s) \sum_w p_w \sum_j T(j \mid s, w) \langle v_j, U_w z_s \rangle \\ &\leq \sum_s \mathbb{P}(S = s) \|z_s\| \max_{\|u\|=1} \sum_w p_w \max_j \langle v_j, U_w u \rangle \leq (1 - D_k(p))q. \end{aligned}$$

Terms with  $z_s = 0$  contribute zero. Replacing an executable assignment by the largest pairing is a relaxation in the correct direction. Arbitrary unobservable hidden coordinates are included in  $s$ ; no statewise observable-vector hypothesis is used. This is the conditional-centroid mechanism of [7, 8], now applied to the optimized finite word datum.  $\square$

For an available profile  $K_\bullet$ , let  $I_0 = 0$  and define

$$(3.5) \quad I_t = \max \left\{ I_{t-1}, \max_{1 \leq B \leq t} [I_{t-B} + \chi_{\mathcal{A}}(K_t, B)] \right\}, \quad \mathcal{I}_N(K_\bullet) = I_N.$$

This is the largest sum of charges of disjoint command intervals, with an interval of length  $B$  ending at  $t$  charged by  $\chi_{\mathcal{A}}(K_t, B)$ . The recurrence depends only on the alphabet and the available profile.

**Theorem 3.3** (A nonuniform interval budget). *Every machine with  $\kappa = \rho/\sqrt{D} - 2\varepsilon > 0$  satisfies*

$$(3.6) \quad \mathcal{I}_N(K_\bullet) \leq \log(1/\kappa).$$

*More generally its optimal actual terminal error obeys*

$$(3.7) \quad \mathcal{E}_N(K_\bullet) \geq \frac{1}{2} \left[ \frac{\rho}{\sqrt{D}} - e^{-\mathcal{I}_N(K_\bullet)} \right]_+.$$

*For a uniform peak bound  $K$ , every  $B \leq N$  gives*

$$(3.8) \quad \lfloor N/B \rfloor \chi_{\mathcal{A}}(K, B) \leq \log(1/\kappa).$$

*For a fixed  $B, k$  with  $\Gamma_{\mathcal{A}}(k, B) > 0$ ,*

$$(3.9) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + \frac{B \log(1/\kappa)}{\chi_{\mathcal{A}}(k, B)}.$$

*If  $\chi = +\infty$ , the last fraction is zero.*

*Proof.* Choose any maximizing disjoint interval family in (3.5). On each selected interval put an independently sampled optimizing complete word from Proposition 3.1. All gaps and the final remainder contain fixed deterministic command words. A deterministic orthogonal command followed by a stochastic update cannot increase the conditional-mean norm, by conditional Jensen. Lemma 3.2 therefore gives

$$q_N \leq \prod_{(t-B, t]} (1 - \Gamma_{\mathcal{A}}(K_t, B)) = e^{-\mathcal{I}_N(K_\bullet)},$$

since  $q_0 = 1$ . The chosen law can depend on the available layer sizes, not on private states. Wordwise correctness requires the same machine to pass it. Lemma 2.2 proves (3.6) and (3.7); taking consecutive intervals proves (3.8).

For (3.9), discard candidate endpoints before  $B$ . Select the earliest remaining endpoint, delete all later candidates less than  $B$  away, and repeat. If  $M$  is the original number and  $J$  the number selected, the explicit count is

$$M - (B - 1) \leq BJ.$$

The associated length- $B$  intervals are disjoint and each contributes at least  $\chi_{\mathcal{A}}(k, B)$ . Applying (3.6) gives the result. No restriction on the other registers was introduced.  $\square$

The cap union estimate below is the geometric certificate already used in the predecessor packet method [7]. The finite incidence formulation specifies how to optimize that certificate on a fully listed word set.

**Proposition 3.4** (Uniform cap certificates). *Suppose a law  $p$  on length- $B$  words satisfies*

$$\sup_{u,v \in \mathbb{S}^{D-1}} p\{w : \|U_w u - v\| < \delta\} \leq b.$$

*Then  $\Gamma_{\mathcal{A}}(k, B) \geq \frac{1}{2}\delta^2(1 - kb)_+$ . For algebraic product matrices and algebraic  $\delta$ , the best such  $b$  is the value of a finite linear program once cap incidence types have been decided by real quantifier elimination.*

*Proof.* The union of  $k$  open caps has probability at most  $kb$ . Outside it,  $1 - \max_j \langle v_j, U_w u \rangle = \frac{1}{2} \min_j \|U_w u - v_j\|^2 \geq \delta^2/2$ . For the algorithmic assertion, a subset of the finite word list either can or cannot be contained in one such cap for some  $u, v$ . This is a finite system of algebraic inequalities, including the sphere equations. For every feasible subset  $A$  impose  $\sum_{w \in A} p_w \leq b$ , along with the simplex constraints, and minimize  $b$ . Every actual incidence set is feasible; conversely a feasible subset is included in an actual incidence set and has no greater mass. Thus these constraints give exactly the desired supremum. The exponentially large enumeration is not concealed.  $\square$

**3.2. A separate exact-synthesis profile.** Fix  $0 < a < 1$ . For  $k$  such that the following family is nonempty, let

$$(3.10) \quad \lambda_{\mathcal{A}}(k, a) = \min_P \inf \{\lambda \geq 1 : U_b P \subseteq \lambda P \text{ for all } b \in \mathcal{A}\},$$

where  $P$  ranges over convex hulls of at most  $k$  points in  $\mathbb{B}^D$  containing  $a\mathbb{B}^D$ . An empty family has value  $+\infty$ .

**Theorem 3.5** (Compatible enclosure synthesis). *The finite value in (3.10) is attained and lies in  $[1, a^{-1}]$ . If  $\rho \lambda_{\mathcal{A}}(k, a)^N \leq a$ , there is an exact machine with  $\max(k, 2)$  labels, common command matrices and a horizon-dependent decoder. Each row has at most  $D + 1$  successors. Algebraic matrices and algebraic  $a$  give an algebraically decidable profile value and algebraic row witnesses. For two indexed alphabets with a common feasible family,*

$$(3.11) \quad |\lambda_U(k, a) - \lambda_V(k, a)| \leq a^{-1} \max_b \|U_b - V_b\|_{\text{op}}.$$

*Proof.* Parameterize  $P$  by  $k$  points in the closed unit ball, allowing repetition. Containment of  $a\mathbb{B}^D$  is closed in these coordinates. Every such  $P$  satisfies  $U_b P \subseteq \mathbb{B}^D \subseteq a^{-1}P$ . The containment conditions are closed on this compact set and  $1 \leq \lambda \leq a^{-1}$ ; hence a minimum exists.

Let its vertices be  $v_s$  and  $\lambda$  its dilation. Initialize with a convex representation of  $ax$ . For each  $b, s$ , represent  $\lambda^{-1}U_b v_s$  by the same vertices and use its convex coefficients as the row  $T_b(s, \cdot)$ . Induction gives

$$\mathbb{E}[v_{S_t} \mid w] = a\lambda^{-t}U_w x.$$

The decoder  $d_j(s) = \rho \lambda^N \langle e_j, v_s \rangle / a$  lies in  $[-1, 1]$  and gives (1.1). These are actual rows throughout the run, not independently chosen checkpoint factorizations. Carathéodory gives  $D + 1$  successors; there are  $|\mathcal{A}|k$  command rows, and their construction and storage remain uncharged.

The defining finite problem is semialgebraic: use convex coefficients for transformed vertices and a universal ball containment condition, then eliminate quantifiers. To prove (3.11), set  $e = \max_b \|U_b - V_b\|_{\text{op}}$ . For every feasible  $P$ ,  $V_b P \subseteq U_b P + e\mathbb{B}^D \subseteq (\lambda + e/a)P$ . Interchanging  $U, V$  completes the proof.  $\square$

The two profiles have different quantifiers. The converse tests every hidden machine; the enclosure theorem builds a particular positive realization. Neither theorem asserts that  $\lambda_{\mathcal{A}}$  is the

optimum over hidden machines. This is why finite-scale lower and upper bounds can be used even when they do not match.

**Proposition 3.6** (Bounded word recoding). *Suppose every command of  $\mathcal{A}$  is a product of exactly  $L$  commands of an alphabet  $\mathcal{B}$  on the same space, with the same seeds and queries. Then*

$$\Gamma_{\mathcal{A}}(k, B) \leq \Gamma_{\mathcal{B}}(k, LB), \quad \lambda_{\mathcal{A}}(k, a) \leq \lambda_{\mathcal{B}}(k, a)^L, \quad W_{N,\varepsilon}(\mathcal{A}) \leq W_{LN,\varepsilon}(\mathcal{B}).$$

*An identity letter permits words of length at most  $L$  to be padded. If both alphabets contain an identity and admit such bounded recodings into the other, the existence and value of a logarithmic width exponent agree.*

*Proof.* Push forward a packet law along the word substitution to obtain the first inequality. Iterate the enclosure containments for the second. For the last inequality, compose the  $L$  stochastic rows of a  $\mathcal{B}$  machine into one row for the corresponding  $\mathcal{A}$  command, retaining only the block-end label. All rows may depend on the external epoch. This is an implementation of the atomic  $\mathcal{A}$  interface, not a claim that its physical letters cost  $L$  letters. Applying the two comparisons and monotonicity under final identity padding proves the exponent statement.  $\square$

**Corollary 3.7** (Packing–dilation balance). *Fix a calibrated experiment, an inradius  $a > \rho$ , and positive numbers  $u, v$ . Suppose that for every sufficiently large  $k$  there are a length  $B(k) \leq Ck^u$  and a word law for which*

$$\sup_{x,z \in \mathbb{S}^{D-1}} p\{w : \|U_w x - z\| < ck^{-v}\} \leq (2k)^{-1}.$$

*Suppose also that  $\lambda_{\mathcal{A}}(k, a) \leq \exp(C'k^{-u-2v})$ . Then*

$$W_{N,\varepsilon} = \Theta(N^{1/(u+2v)}).$$

*The same conclusion holds with  $u = 0$  when the lengths are uniformly bounded.*

*Proof.* Proposition 3.4 gives  $\Gamma(k, B(k)) \geq c^2 k^{-2v}/4$ . Equation (3.8) implies  $N \leq C_1 k^{u+2v}$  for a feasible peak  $k$ . Conversely choose  $k$  at a sufficiently large constant times  $N^{1/(u+2v)}$  so that  $C'Nk^{-u-2v} \leq \log(a/\rho)$ , and apply Theorem 3.5. The hypotheses concern finite word orbits and compatible convex containments, and impose no commutativity on the matrices.  $\square$

#### 4. EXACT OPTIMIZATION OF PLANAR PACKETS

For a rotation alphabet, let  $z_1, \dots, z_M \in \mathbb{S}^1 \subset \mathbb{C}$  be the distinct product phases of words of length  $B$ , ordered cyclically. A common rotation of all phases can be absorbed into the decoder centers. The initial direction in (3.1) therefore disappears. Any probability distribution on these phases is implemented by a distribution on representative words of the same length.

Let  $\mathfrak{P}_k$  be the finite set of partitions of the cyclically ordered list into at most  $k$  nonempty consecutive arcs. For  $p \in \Delta_M$ , set

$$(4.1) \quad F_k(p) = \max_{\mathcal{P} \in \mathfrak{P}_k} \sum_{A \in \mathcal{P}} \left| \sum_{i \in A} p_i z_i \right|.$$

**Theorem 4.1** (Optimal word laws as a cone program). *For the planar alphabet above,*

$$(4.2) \quad \Gamma_{\mathcal{A}}(k, B) = 1 - \min_{p \in \Delta_M} F_k(p).$$

*Equivalently, minimize  $t$  subject to  $p \in \Delta_M$  and, for every  $\mathcal{P} \in \mathfrak{P}_k$ ,*

$$(4.3) \quad \left\| \sum_{i \in A} p_i (\Re z_i, \Im z_i) \right\|_2 \leq r_{\mathcal{P},A} \quad (A \in \mathcal{P}), \quad \sum_{A \in \mathcal{P}} r_{\mathcal{P},A} \leq t.$$

*This is a finite second-order cone program with attained optimum. A fixed law can be evaluated by cyclic segmentation in  $O(kM^3)$  arithmetic operations, including computation of all interval sums. The optimization over laws may have exponentially many partition constraints.*

*Proof.* For fixed centers  $v_1, \dots, v_k$  assign each phase to a nearest center in chordal distance; equivalently maximize  $\Re(\bar{v}_j z_i)$ . Distinct centers have connected cyclic Voronoi arcs, bounded by the angular midpoints of adjacent centers. Ties can be assigned consecutively; coincident centers can be merged. For the induced partition, the contribution of a block  $A$  is at most  $|\sum_{i \in A} p_i z_i|$ .

Conversely, for any cyclic partition choose the center of each block in the direction of its weighted complex sum, arbitrarily if that sum vanishes. The best assignment to those centers has value at least the sum in (4.1). Thus the maximum over centers equals  $F_k(p)$ . This proves (4.2); introducing the norm epigraphs proves (4.3). Zero weights follow either directly or by continuity from positive weights.

For a fixed cyclic cut, prefix sums give every block vector in constant time. The usual recurrence for the best  $j$  blocks among the first  $m$  points checks at most  $M$  previous split positions. It costs  $O(kM^2)$  for that cut. Repeating for the  $M$  cuts costs  $O(kM^3)$ . This recurrence evaluates a given law; it is not a polynomial-size representation of all constraints in (4.3). Compactness of the simplex and continuity give attainment.  $\square$

Nearest-center partitions and vector quantization are classical [5]. The additional optimization here chooses the distribution of actual command words, and Theorem 3.3 turns its value into a constraint on arbitrary hidden registers. There is no assertion that static quantization alone characterizes the least whole-run width.

**Proposition 4.2** (A finite exact packet certificate). *For the alphabet consisting of identity and a quarter turn, at word length three the four distinct phases are  $1, i, -1, -i$ . Their uniform law is optimal and*

$$\Gamma(1, 3) = 1, \quad \Gamma(2, 3) = 1 - \frac{1}{\sqrt{2}}, \quad \Gamma(3, 3) = \frac{2 - \sqrt{2}}{4}, \quad \Gamma(k, 3) = 0 \quad (k \geq 4).$$

*Proof.* The distortion is concave in the phase law and invariant under its four cyclic rotations. Averaging any law over those rotations proves optimality of the uniform law. For two centers, adjacent pairs give total  $\sqrt{2}/2$ ; a triple and a singleton give only  $1/2$ . For three centers, one pair and two singletons give  $\sqrt{2}/4 + 1/2$ . For one center the sum is zero; four centers fit all phases. Formula (4.2) gives the values. A uniform phase law is realized by the four ordinary words with zero, one, two or three initial quarter turns and then idle commands.  $\square$

For algebraic rotation matrices, both the general quantifier-elimination formulation and the cone program have algebraic coefficients. No finite list of spherical harmonics is used. For merely computable-real entries, finite numerical approximations are diagnostics unless a rigorous error enclosure is supplied; (3.3) then quantifies how an entry enclosure propagates to a packet certificate.

**Corollary 4.3** (Separated phases). *If  $M \geq 2k$  distinct length- $B$  word phases have pairwise circular distance at least  $s \in (0, 1/2]$ , then  $\Gamma_A(k, B) \geq s^2$ .*

*Proof.* Use their uniform law. Each open arc of circular radius  $s/2$  contains at most one phase. At least half the mass is at distance at least  $s/2$  from every one of  $k$  centers. On  $[0, 1/2]$ ,  $1 - \cos(2\pi t) = 2\sin^2(\pi t) \geq 8t^2$ . The average loss is at least  $s^2$ . This is a lower certificate for (4.2), not necessarily its exact value.  $\square$

## 5. ARITHMETIC PROFILES AT EVERY FINITE SCALE

Now  $\mathcal{A}_\alpha = \{I, R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_r}\}$ , with  $r$  fixed. Define

$$(5.1) \quad \psi_\alpha(H) = \min_{0 < \|m\|_1 \leq H} \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}, \quad \eta_\alpha(q) = \max_{1 \leq i \leq r} \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}.$$

These are finite functions of the alphabet parameters. The first may vanish in the presence of a rational relation. The upper profile remains usable in that case.

**Theorem 5.1** (Finite arithmetic bounds). *Let  $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$ . For a machine of peak  $K$ , put  $L = \lceil (2K)^{1/r} \rceil$  and  $B = r(L - 1)$ . Then*

$$(5.2) \quad \lfloor N/B \rfloor \psi_\alpha(B)^2 \leq \log(1/\kappa).$$

For every integer  $q \geq 4$ ,

$$(5.3) \quad N\eta_\alpha(q) \leq q^2/100 \implies W_{N,0}(\alpha) \leq q.$$

Consequently every finite planar rotation alphabet satisfies

$$(5.4) \quad W_{N,0}(\alpha) \leq \max(4, \lceil \sqrt{50N} \rceil).$$

All upper bounds use common command rows and hold also after adjoining reflection in the first coordinate axis.

*Proof.* Construct a word of length  $B$  in  $r$  consecutive segments of  $L - 1$  positions. In segment  $i$  give  $J_i$  copies of the  $i$ th rotation followed by idle commands, independently and uniformly for  $J_i \in \{0, \dots, L - 1\}$ . Differences of count vectors have  $\ell_1$  norm at most  $B$ . If  $\psi_\alpha(B) > 0$ , the  $L^r \geq 2K$  phases are distinct and separated by  $\psi_\alpha(B)$ . Corollary 4.3 and (3.8), with  $-\log(1-x) \geq x$ , give (5.2). If the minimum is zero the displayed inequality is automatic. All letters in a segment are actual commands, not a free count input.

For the upper bound let  $P_q$  be the regular  $q$ -gon inscribed in the unit circle. Choose nearest integers  $p_i$  to  $q\alpha_i$ , and put  $\delta_i = 2\pi(\alpha_i - p_i/q)$ , so  $|\delta_i| \leq \pi/q$ . Elementary chord geometry gives

$$(5.5) \quad R_{2\pi\alpha_i}P_q \subseteq \Lambda_i P_q, \quad \Lambda_i = \frac{\cos(\pi/q - |\delta_i|)}{\cos(\pi/q)}, \quad \log \Lambda_i \leq \frac{4\pi^2\eta_\alpha(q)}{q^2}.$$

Indeed the ray of a rotated vertex meets a neighboring edge at radial coordinate  $\cos(\pi/q)/\cos(\pi/q - |\delta_i|)$ . Integrating the derivative of  $\log \cos(\pi/q - t)$  bounds the logarithm by  $|\delta_i| \tan(\pi/q)$ , and  $\tan(\pi/q) \leq 2\pi/q$  for  $q \geq 4$ .

Take  $\Lambda = \max_i \Lambda_i$ , including one for the identity. Under (5.3),  $\Lambda^N \leq e^{\pi^2/25} < 2$ . Since  $P_q$  contains the half-unit disk, Theorem 3.5 with  $a = 1/2$  applies:  $\rho\Lambda^N \leq 1/5 < 1/2$ . Explicitly, initialize at mean  $x/2$ , represent  $\Lambda^{-1}R_{2\pi\alpha_i}v_s$  by the vertices for each row, and decode by  $2\rho\Lambda^N \langle e_j, v_s \rangle$ . This verifies every numerical word, including any deterministic remainder of a converse packet decomposition.

There are  $(r + 1)q$  common command rows and two decoder columns, each row having at most three successors. These counts exclude row-description and arithmetic costs. Since  $\eta_\alpha(q) \leq 1/2$ , (5.4) follows. Reflection preserves  $P_q$ ; its contracted image uses the same vertex set. The reflected lower bound is obtained simply by restricting to rotation-only words.  $\square$

The pair  $(\psi_\alpha, \eta_\alpha)$  is a specialization of the word-distortion and enclosure profiles. It is not assumed to characterize exact finite width. In particular, (5.2) can be weak when long word packets have clusters; the optimized profile of Section 4 is available without that separation relaxation.

**Lemma 5.2** (Denominators under a general power bound). *Suppose for  $\tau \geq r$  and  $b > 0$ ,*

$$(5.6) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b \|m\|_1^{-\tau} \quad (m \neq 0).$$

*For every integer  $Q \geq 2$  there is  $q$  with*

$$(5.7) \quad cQ^{r/(\tau+1-r)} \leq q \leq Q^r, \quad \eta_\alpha(q) \leq Q^{-1}, \quad c = (b/r^{\tau+1})^{r/(\tau+1-r)}.$$

*Proof.* The  $Q^r + 1$  points  $j\alpha$  in  $Q^r$  equal torus boxes give integers  $p_i, q$  with  $1 \leq q \leq Q^r$  and  $|q\alpha_i - p_i| \leq Q^{-1}$ . For  $M = \lfloor q^{1/r} \rfloor$ , the  $(M+1)^r > q$  integer vectors in  $\{0, \dots, M\}^r$  have a residue collision for  $m \cdot p \pmod{q}$ . Subtracting gives  $m \neq 0$  with  $H = \|m\|_1 \leq rq^{1/r}$  and  $q \mid m \cdot p$ . Thus

$$bH^{-\tau} \leq \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad bqQ \leq r^{\tau+1}q^{(\tau+1)/r}.$$

Rearranging yields (5.7). This direct residue argument supplies exactly the quantitative transference needed here, without an unspecified transference constant.  $\square$

**Corollary 5.3** (Power bounds at finite Diophantine type). *Under (5.6), for constants depending on the fixed data,*

$$(5.8) \quad c_1 N^{r/(2\tau+1)} \leq W_{N,\varepsilon}(\alpha) \leq C_1 N^{\min\{1/2, r(\tau+1-r)/(\tau+1+r)\}}.$$

*No matching claim is made when  $\tau > r$ .*

*Proof.* From (5.2),  $N < B + b^{-2}B^{2\tau+1} \log(1/\kappa)$ , while  $B \leq r(2K)^{1/r}$ . This gives the lower power after adjusting finitely many initial horizons. Lemma 5.2 bounds  $N\eta_\alpha(q)/q^2 \leq c^{-2}NQ^{-1-2r/(\tau+1-r)}$ . Choose  $Q$  at constant times  $N^{(\tau+1-r)/(\tau+1+r)}$ , large enough to satisfy (5.3) and  $q \geq 4$ . Then  $q \leq Q^r$  proves the second candidate upper power; (5.4) proves the first.  $\square$

**Corollary 5.4** (The retained badly approximable law). *If (5.6) holds with  $\tau = r$ , then  $W_{N,\varepsilon}(\alpha) = \Theta(N^{r/(2r+1)})$ , also with the reflection extension. This is the previously established matched law [8].*

*Proof.* Substitute  $\tau = r$  in (5.8); the two powers coincide.  $\square$

For  $1, \alpha_1, \dots, \alpha_r$  rationally independent, every optimized finite-block action gap on the circle is zero. Indeed simultaneous approximation supplies unbounded frequencies  $n$  with every  $n\alpha_i$  close to an integer. Each fixed rotation word then acts almost identically on its Fourier mode. For the reflected alphabet use the real cosine mode, which reflection fixes. This is the established almost-invariance argument, not a claim that the finite packet distortion vanishes.

## 6. MINIMAL ORDINARY TYPE AND ALMOST-EVERYWHERE WIDTHS

Uniform bad approximation is substantially stronger than the condition needed to determine the logarithmic width exponent. We first deduce the minimal-type assertion without any measure-theoretic hypothesis.

*Proof of (1.3) in Theorem 1.1.* If  $\omega^*(\alpha) = r$ , then for every  $h > 0$  there is  $b_h > 0$  such that

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b_h \|m\|_1^{-r-h} \quad (m \neq 0).$$

Indeed the definition of the limsup gives this inequality for all sufficiently large  $\|m\|_1$ , and rational independence makes the finite remaining minimum positive. Apply Corollary 5.3 with  $\tau = r + h$ . It gives

$$\frac{r}{2r+2h+1} \leq \liminf_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \limsup_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \frac{r(1+h)}{2r+1+h}.$$

Let  $h \downarrow 0$ . This proves (1.3). No positive uniform constant at exponent  $r$  was used.  $\square$

For completeness we prove a more quantitative metric statement, using only the summable part of metric Diophantine approximation. It does not claim a new metric theorem.

**Lemma 6.1** (A summable logarithmic separation). *For Lebesgue-almost every  $\alpha \in [0, 1]^r$ , and every  $s > 1$ , there is  $b_s > 0$  such that*

$$(6.1) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{b_s}{\|m\|_1^r (\log(2 + \|m\|_1))^s} \quad (m \neq 0).$$

Such a vector has  $\omega^*(\alpha) = r$ .

*Proof.* For a nonzero integer vector  $m$ , the map  $\alpha \mapsto m \cdot \alpha \pmod{1}$  sends Haar measure on the torus to uniform circle measure. There are at most  $C_r H^{r-1}$  integer vectors with  $\|m\|_1 = H$ . Hence the measures of the events

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} < [\|m\|_1^r (\log(2 + \|m\|_1))^s]^{-1}$$

have finite sum, bounded by a constant times  $\sum_{H \geq 1} [H(\log(2 + H))^s]^{-1}$ . The measure of their tail union tends to zero. Almost every vector violates the unscaled inequality only finitely often. Remove the measure-zero rational hyperplanes and reduce the constant to cover the finite exceptions. Intersecting the full-measure sets for rational  $s > 1$  gives the assertion for every real  $s > 1$ .

This bound implies  $\omega^*(\alpha) \leq r$ . For the reverse inequality, place the  $(M + 1)^r$  values  $m \cdot \alpha \pmod{1}$ ,  $m \in \{0, \dots, M\}^r$ , in fewer than  $(M + 1)^r$  intervals of equal length. A collision gives a nonzero difference of norm at most  $rM$  and distance  $O(M^{-r})$ . As  $M \rightarrow \infty$  the norms of some such differences must tend to infinity, since rational independence excludes an exact relation. Thus  $\omega^*(\alpha) \geq r$ .  $\square$

**Proposition 6.2** (Quantitative bounds under logarithmic separation). *Suppose (6.1) holds for some  $s > 0$  and  $b_s > 0$ . Then there are  $c, C > 0$  such that*

$$(6.2) \quad cN^{r/(2r+1)}(\log(N+2))^{-2rs/(2r+1)} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)}(\log(N+2))^{2r^2s/(2r+1)}.$$

*Proof.* The packet bound (5.2) gives

$$N < B + b_s^{-2} B^{2r+1} (\log(2 + B))^{2s} \log(1/\kappa).$$

With  $B \leq r(2K)^{1/r}$  this implies  $N \leq C_0 K^{(2r+1)/r} (\log(2 + K))^{2s}$ . If  $K > N$ , the proposed lower bound is already true after reducing its constant. Otherwise  $\log(2 + K) \leq \log(2 + N)$ , and rearrangement proves the first inequality of (6.2).

For the upper inequality repeat the residue argument of Lemma 5.2. It produces  $q \leq Q^r$ ,  $\eta_\alpha(q) \leq Q^{-1}$  and  $H \leq rq^{1/r}$  with

$$\frac{b_s}{H^r (\log(2 + H))^s} \leq \frac{H}{qQ}.$$

Since  $q \leq Q^r$ , it follows for  $Q \geq 2$  that

$$(6.3) \quad q \geq c_0 \frac{Q^r}{(\log(Q+2))^{rs}}.$$

Indeed  $b_s q Q \leq H^{r+1} (\log(2 + H))^s \leq C_r q^{1+1/r} (\log(Q+2))^s$ , and raising the rearranged inequality to the  $r$ th power gives (6.3).

Choose an integer

$$Q = \left\lceil C_2 N^{1/(2r+1)} (\log(N+2))^{2rs/(2r+1)} \right\rceil.$$

For sufficiently large fixed  $C_2$ , (6.3) implies  $N\eta_\alpha(q)/q^2 \leq 1/100$  and  $q \geq 4$ . To see the constant choice explicitly, for large  $N$  one has  $\log(Q+2) \leq C_3 \log(N+2)$  with  $C_3$  fixed; thus the ratio is bounded by a constant times  $C_2^{-(2r+1)}$ . Enlarge the final constant for the finitely many initial horizons. Theorem 5.1 gives an exact machine with  $q \leq Q^r$  labels, proving the upper bound.  $\square$

*Completion of Theorem 1.1.* Apply Lemma 6.1 with  $s = 1 + \delta$  and then Proposition 6.2. This yields (1.4) on one full-measure set for all fixed  $\delta > 0$ . These statements concern a fixed alphabet before the horizon grows. None of their constants is asserted uniform in  $r$  or near a rational relation.  $\square$

**6.1. Rational dependencies without redundant exponents.** The number of listed rotations need not be the relevant arithmetic rank. The following result covers finite rational reparametrizations rather than assigning one exponent to every redundant letter.

**Corollary 6.3** (Finite rational extensions). *Suppose an alphabet contains  $I$  and rotations through angles  $\beta_1, \dots, \beta_d$ , where  $1, \beta_1, \dots, \beta_d$  are rationally independent and  $\omega^*(\beta) = d$ . Suppose every additional angle belongs to  $\mathbb{Q} + \sum_{j=1}^d \mathbb{Q}\beta_j$ . Then*

$$\lim_N \frac{\log W_{N,\varepsilon}}{\log N} = \frac{d}{2d+1}.$$

*The same assertion holds on a full-measure set of  $\beta$  with the logarithmic upper and lower powers of (1.4), replacing  $r$  by  $d$ . Reflection may be added. If all angles are rational, exact width is bounded independently of  $N$ .*

*Proof.* Restriction to the displayed  $d$ -angle subalphabet gives the lower bound. Choose one common denominator  $h$  and integers  $t_i, n_{ij}$  so that every angle is  $t_i/h + \sum_j n_{ij}\gamma_j$  with  $\gamma_j = \beta_j/h$ . Multiplication by  $h$  shows  $\|m \cdot \gamma\|_{\mathbb{R}/\mathbb{Z}} \geq h^{-1} \|m \cdot \beta\|_{\mathbb{R}/\mathbb{Z}}$ . Thus every separating power or logarithmic inequality for  $\beta$  holds for  $\gamma$  with a changed constant.

Take a simultaneous approximant  $p_j/q$  to  $\gamma$  from the proof above. A command's irrational part has residual angle at most  $2\pi(\sum_j |n_{ij}|)/(qQ)$ . The same polygon construction therefore works, after changing only fixed constants. Store in addition the rational phase  $c \in \mathbb{Z}/h\mathbb{Z}$ , updated by  $c \mapsto c + t_i$ . The decoder rotates the stored polygon vector by  $2\pi c/h$ . This uses  $hq$  labels, not an uncharged phase. Reflection sends  $c$  to  $-c$  and conjugates the polygon phase; the regular polygon and the construction are reflection-invariant. The exact expectation identity is unchanged. The finite multiplier  $h$  does not change a logarithmic exponent or the displayed logarithmic powers.

For a purely rational alphabet, choose a common period  $h$  and retain the seed identifier and the accumulated phase. This uses at most  $4h$  command labels for our four planar seeds, plus the held binary answer convention. No infinite-gap hypothesis is involved.  $\square$

## 7. EVERY IRRATIONAL ANGLE HAS CUBIC SCALES

We keep the two-letter alphabet  $\{I, R_{2\pi\alpha}\}$  and the planar resource and error conventions. No Diophantine lower bound is assumed in this section. The needed continued-fraction estimates are classical [3]; we include their short proof to specify the exact finite inequalities being used.

**Lemma 7.1** (Convergent separation). *Let  $p_n/q_n$  be the convergents of an irrational  $\alpha$ , ignoring the finitely many initial indices with repeated denominator. Then*

$$(7.1) \quad \frac{1}{q_{n+1} + q_n} < |q_n\alpha - p_n| < \frac{1}{q_{n+1}}, \quad \min_{1 \leq l < q_n} \|l\alpha\|_{\mathbb{R}/\mathbb{Z}} > \frac{1}{q_n + q_{n-1}} > \frac{1}{2q_n}.$$

*Proof.* The convergent recurrences imply  $p_n q_{n-1} - p_{n-1} q_n = \pm 1$  and alternating signs of  $e_n = q_n \alpha - p_n$ . If  $\xi$  is the next complete quotient, then

$$\alpha = \frac{p_n \xi + p_{n-1}}{q_n \xi + q_{n-1}}, \quad a_{n+1} < \xi < a_{n+1} + 1.$$

It follows that  $|e_n| = 1/(q_n \xi + q_{n-1})$ , proving the first pair.

For  $0 < l < q_n$  and an integer  $p$ , the determinant identity writes  $(p, l) = a(p_n, q_n) + b(p_{n-1}, q_{n-1})$  with integers  $a, b$ . Here  $b \neq 0$ ;  $a, b$  cannot both be positive, and cannot both be negative. If  $a = 0$

the error has modulus at least  $|e_{n-1}|$ . Otherwise they have opposite signs, so  $ae_n$  and  $be_{n-1}$  have the same sign and again  $|l\alpha - p| \geq |e_{n-1}|$ . Minimize over  $p$  and apply the first pair with index  $n - 1$ . This proves the remaining inequalities.  $\square$

**Theorem 7.2** (Universal cubic subsequences). *Set  $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$ ,  $C_\kappa = 8(1 + \log(1/\kappa))$ , and*

$$M_n = \lceil C_\kappa q_n^3 \rceil, \quad m_\kappa = \lceil 100(C_\kappa + 1) \rceil.$$

*For all sufficiently large  $n$ ,*

$$(7.2) \quad q_n/2 < W_{M_n, \varepsilon}(\alpha) \leq m_\kappa q_n.$$

*The upper bound is exact, uses common command rows and requires no regularity of the partial quotients.*

*Proof.* Use words of length  $B = q_n - 1$  whose number of initial active letters ranges uniformly over  $0, \dots, q_n - 1$ . Lemma 7.1 gives  $q_n$  distinct phases with separation greater than  $1/(2q_n)$ . If the proposed peak  $K$  were at most  $q_n/2$ , Corollary 4.3 and the packet budget would give

$$\left\lfloor \frac{M_n}{q_n - 1} \right\rfloor \frac{1}{4q_n^2} \leq \log(1/\kappa).$$

The left side is at least  $C_\kappa/4 - 1/(4q_n^2)$ , strictly greater than the right side. This proves the lower bound against all hidden-state machines.

For the upper bound use the polygon denominator  $Q_n = m_\kappa q_n$ . Distance to the nearest integer is subadditive, so

$$\|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq m_\kappa |q_n \alpha - p_n| < m_\kappa / q_{n+1} \leq m_\kappa / q_n.$$

Consequently  $Q_n^2 / [100 \|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}}] > m_\kappa q_n^3 / 100 \geq (C_\kappa + 1) q_n^3 \geq M_n$ . Equation (5.3) gives the exact upper construction. The constants have been chosen for transparency, not optimality.  $\square$

**Proposition 7.3** (Resonant long horizons). *For any irrational  $\alpha$  and integer  $q \geq 4$  with  $N_q = \lfloor q^2 / (100 \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}) \rfloor \geq 1$ , one has  $W_{N_q, 0}(\alpha) \leq q$ . If  $\alpha$  is Liouville, there is a sequence  $N \rightarrow \infty$  on which  $\log W_{N, \varepsilon}(\alpha) / \log N \rightarrow 0$ .*

*Proof.* The first assertion is the finite enclosure implication (5.3). For a Liouville number choose arbitrarily large  $A$  and unbounded denominators  $q$  with  $0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < q^{-A}$ . The first assertion then supplies horizons  $N_q \geq q^{A+2}/200$  for large  $q$ , with width at most  $q$ . The logarithmic ratio is at most  $1/(A+2) + o(1)$ . A diagonal sequence with  $A \rightarrow \infty$  proves the claim, since widths are at least two.  $\square$

*Proof of Theorem 1.2.* Equation (7.2) and  $M_n = \Theta(q_n^3)$  give the first assertion and a subsequence of logarithmic exponent  $1/3$ . Proposition 7.3 gives the zero lower limit for Liouville angles, and (5.4) gives the upper limit at most  $1/2$ .

It remains to verify unboundedness, which does not follow from a subsequence alone. Fix any proposed width  $K$ . The  $2K$  phases of counts  $0, \dots, 2K - 1$  are distinct for an irrational angle and have some positive minimum separation  $s_K$ . Independent words of length  $2K - 1$  imply  $\lfloor N/(2K - 1) \rfloor s_K^2 \leq \log(1/\kappa)$  for a width- $K$  machine. This fails for all sufficiently large  $N$ . Thus  $W_{N, \varepsilon} \rightarrow \infty$ .  $\square$

**Corollary 7.4** (An explicit fluctuating alphabet). *Let  $\alpha_L = \sum_{j=1}^{\infty} 10^{-j!}$ . For  $q_m = 10^{m!}$  and all large  $m$ , the horizon*

$$N_m = \lfloor q_m^{m+2} / 200 \rfloor$$

*admits an exact machine with at most  $q_m$  labels. The same fixed two-letter alphabet also has the cubic subsequences of Theorem 7.2; hence no single logarithmic width exponent exists. Adding reflection preserves these conclusions.*

*Proof.* The first  $m$  summands have denominator dividing  $q_m$ , and their remaining positive tail is less than  $2 \cdot 10^{-(m+1)!}$ . Thus  $0 < \|q_m \alpha_L\|_{\mathbb{R}/\mathbb{Z}} < 2q_m^{-m}$  for large  $m$ , proving the horizon bound by (5.3). The same tail estimates preclude rationality and show the Liouville property. The general cubic theorem applies. Reflection leaves the upper polygons invariant, while the rotation-only test words give the same lower bounds.  $\square$

These theorems describe different scales of one and the same experiment. A small label count on a long resonant horizon is not a constant-width simulator for all horizons, and it need not persist across the gaps between successive approximants. The exact finite optimum and the Liouville upper logarithmic limit remain separate quantitative questions. None of the proofs substitutes approximate output for the exact upper realizations.

## 8. ARITHMETIC, QUANTIZATION AND NUMERICAL SEMANTICS

Kronecker quasi-uniformity. Dick–Goda–Larcher–Pillichshammer–Suzuki [4], Theorem 3.9, characterize quasi-uniformity of the multidimensional sequence  $n\alpha \pmod{\mathbb{Z}^r}$  by the simultaneous bound  $\eta_\alpha(q) \geq cq^{-1/r}$ . Their Lemma 3.11 bounds ratios of successive best simultaneous denominators; Section 3.3 also contains the vector  $(2^{1/(r+1)}, \dots, 2^{r/(r+1)})$ . These are direct arithmetic antecedents, not new examples of this paper.

Our packet points instead have the form  $m \cdot \alpha \pmod{1}$ . The relevant separation is dual. The residue argument of Lemma 5.2, with an arbitrary nearest-integer numerator and  $\eta_\alpha(q)$  in place of  $Q^{-1}$ , proves directly that a dual bound with exponent  $r$  implies

$$\eta_\alpha(q) \geq br^{-r-1}q^{-1/r}.$$

Thus the cited quasi-uniformity theorem applies to the uniform badly approximable class used in the predecessor. Its denominator control supplies the arithmetic scales for a polygon upper bound. It supplies neither a compatible stochastic transition table nor a lower bound against discarded hidden history. The logarithmic separation of Lemma 6.1 is weaker than a uniform badly approximable inequality; it is proved by a summable tail estimate, not asserted to imply quasi-uniformity.

What is dynamic in the paired profiles. The geometry of nearest-center distortion and convex hull approximation is classical [5, 11]. A static  $k$ -center factorization does not specify how its encoder is calculated without retaining discarded inputs. Conversely, a convex codebook does not force arbitrary hidden states to possess coordinates in that codebook. The two implications in Section 3 avoid both identifications: the converse conditions on the actual state, and the upper theorem provides the same legal rows throughout the run. The packing–dilation balance in Corollary 3.7 is a conditional theorem on independently specified finite profiles, not a claimed equality of static and dynamic optima.

The circle program makes one of these profiles exactly finite, whereas the all-harmonic action-gap optimization of the predecessor remains infinite-dimensional. A finite word length can be exhaustively listed; an unbounded harmonic degree cannot be certified by a finite truncation without a tail theorem. No general action-gap algorithm follows from Proposition 3.1.

Branching programs and probability preservation. At fixed horizon our machine is a probabilistic ordered read-once layered program with arbitrary stochastic rows and bounded real output columns. We distinguish width from total size, uniform description and random-bit complexity. Fourier growth for regular, permutation and width-three branching programs concerns different hypotheses and objectives [9, 10, 6]. Circle-walk harmonic damping and its arithmetic behavior are also established subjects [2]. Our converse can use correlated letters inside independently sampled complete packets; it is not a theorem about average error under fair independent letters.

Preserving a strict cutpoint sign is weaker than preserving a numerical probability. The predecessor’s constant-width sign construction contracts the output amplitude with length and

hence does not contradict the present width growth. Here signal and tolerance are fixed, one query is selected after the word, and neither joint answers nor anytime correctness is required. Exact upper rows in this article are atomic real kernels, not finite-fair-bit algorithms. A row with three successors can still have costly or irrational coefficients.

Conclusions. The finite optimized word profile and the compatible enclosure profile retain quantitative information which qualitative gap data discard. They yield a minimal-type exponent and an almost-everywhere logarithmic estimate, and they exhibit fluctuating numerical width for every Liouville single angle. The reflected versions show that these conclusions survive a particular noncommuting finite extension, without asserting a classification of arbitrary noncommuting nongapped alphabets.

Exact finite widths, the Liouville upper logarithmic limit, optimal enclosure shapes and a general comparison between the two profiles remain distinct quantitative problems. The results here do not depend on a new external spectral-gap computation. The earlier expansion, calibration and representation theorems retain their original hypotheses in the unchanged supporting article.

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